

Manufacturing Processes — Overview

Manufacturing is the process of converting raw materials into finished products through various techniques. The choice of manufacturing process depends on the material, part geometry, production volume, required tolerances, and cost constraints.

Major categories of manufacturing processes:

- Casting and Molding
- Forming and Shaping (forging, rolling, extrusion)
- Machining (turning, milling, drilling)
- Joining (welding, brazing, soldering)
- Additive Manufacturing (3D printing)

Casting Processes

Casting is one of the oldest manufacturing processes. Molten metal is poured into a mold cavity and allowed to solidify into the desired shape.

Types of casting:

- Sand Casting: uses a sand mold; suitable for large parts and low production volumes.
- Die Casting: uses a metal mold under high pressure; ideal for high volume production of non-ferrous metals.
- Investment Casting (Lost Wax): produces parts with excellent surface finish and dimensional accuracy.

Advantages: complex shapes, large parts, wide range of alloys.

Disadvantages: porosity defects, limited dimensional accuracy (in sand casting), high tooling cost (in die casting).

Forging and Machining

Forging is a forming process in which compressive forces are applied to a workpiece to shape it. Forging improves the grain structure of the metal, resulting in superior mechanical properties compared to casting.

- Open-Die Forging: the workpiece is compressed between flat dies.
- Closed-Die (Impression) Forging: the workpiece is shaped within a die cavity; better dimensional accuracy.

Machining removes material from a workpiece using cutting tools.

Common operations:

- Turning (lathe): produces cylindrical parts.
- Milling: uses a rotating cutter to remove material.
- Drilling: creates holes.

Machining offers excellent dimensional accuracy and surface finish but generates waste material (chips).